

❑ SLIDE: WAVEFORM DESIGN CHEAT SHEET

Mathematical Signal Generation via JavaScript Engine

Core Formula Architecture:

`Offset + (Amplitude * Math.function(t * Frequency * 2 * Math.PI + Phase))`

(Note: t represents continuous time in seconds)

Signal Profile	Mathematical Concept	JavaScript Formula Example	Expected Output Visualization
Pure Sine	Standard oscillation	<code>Math.sin(t * 3) * 5</code>	Smooth, continuous rolling wave
Square	Polarity extraction	<code>Math.sign(Math.sin(t * 3)) * 5</code>	Blocky, alternating digital pulses
Triangle	Linear inverse sine	<code>Math.asin(Math.sin(t*3)) * (2/Math.PI) * 5</code>	Sharp peaks with straight linear slopes
Sawtooth	Modulo time division	<code>((t * 5) % 2) - 1</code>	Steady ramp up, instantaneous snap down
Full-Wave Rectified	Absolute value constraint	<code>Math.abs(Math.sin(t * 3) * 5)</code>	Continuous bouncing positive arches
Half-Wave Rectified	Zero-clamped floor	<code>Math.max(0, Math.sin(t * 3) * 5)</code>	Alternating arch and flatline
Random	Stochastic	<code>(Math.random() * 10) - 5</code>	Scattered, high-

Signal Profile	Mathematical Concept	JavaScript Formula Example	Expected Output Visualization
Noise	generation		frequency static
Noisy Sine	Signal + Noise additive	<code>(Math.sin(t*3)*5) + (Math.random()*2-1)</code>	Fundamental sine wave with jagged edges
Amplitude Modulation	Frequency multiplication	<code>Math.sin(t * 15) * Math.sin(t * 1) * 5</code>	High-speed carrier in expanding/shrinking envelope
Harmonic Synthesis	Multi-frequency addition	<code>(Math.sin(t*2)*5) + (Math.sin(t*6)*1.5)</code>	Complex, repeating ripple structures